

87654321

1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.

2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.

3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

REV

ECN

DESCRIPTION OF REVISION

CK APPD

2011-01-24

CHINA

FDX

SCHEM, RIO, D2

PIB2B, 03/16/2012

820-3071-10.brd

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07/01/2011

10/11/2010

07/01/2011

07/07/2011

06/25/2010

05/14/2010

ALIASES

RESOLVED

Schematic / PCB #'s

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
051-8974	1	SCHEM, RIO, J5	SCH	CRITICAL	
820-3071	1	PCBF, RIO, J5	PCB	CRITICAL	

DRAWING

TITLE=TITLE

ABBREV=DRAWING

LAST_MODIFIED=FY1 Mar 16 13:44:38 2012

SCHEM, RIO, D2

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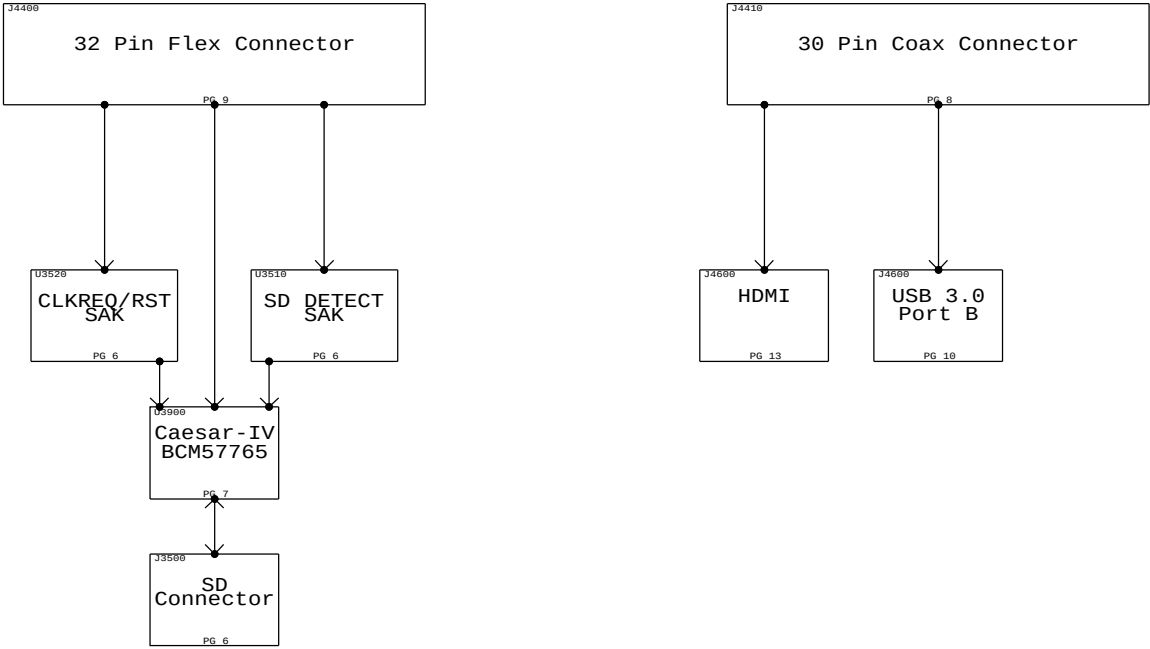
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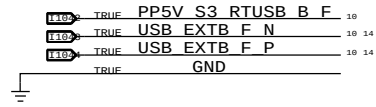
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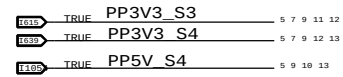


Functional Test Points

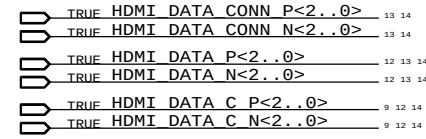
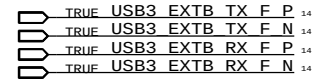
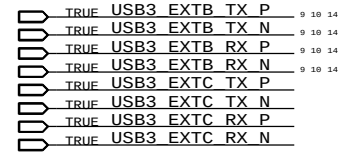
USB PORTS

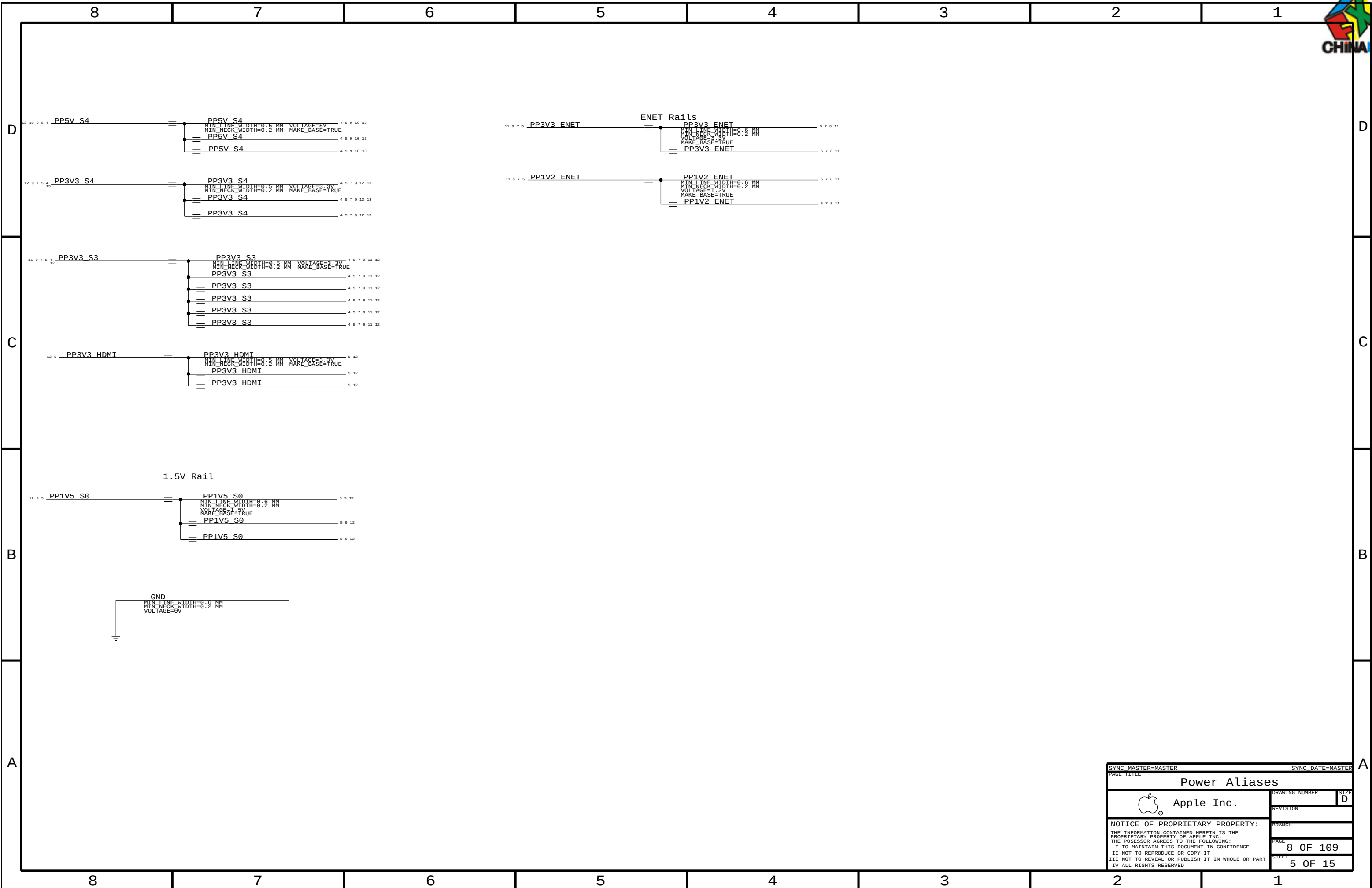


POWER RAILS



NO_TEST=TRUE







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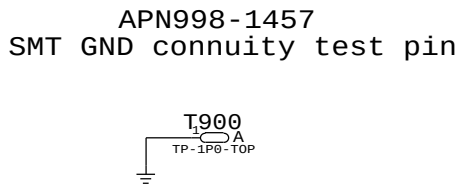
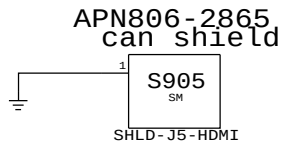
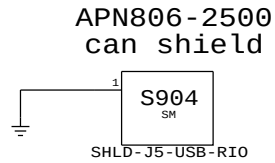
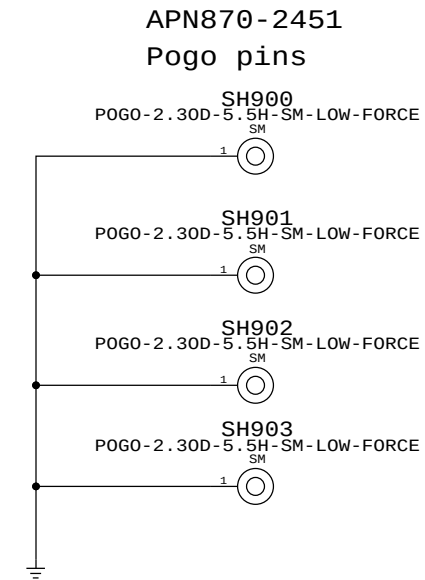
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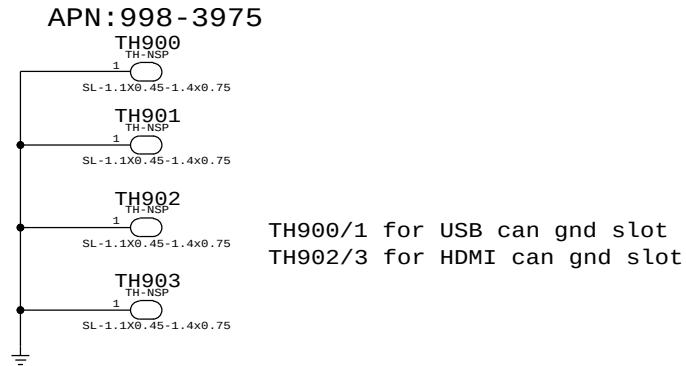
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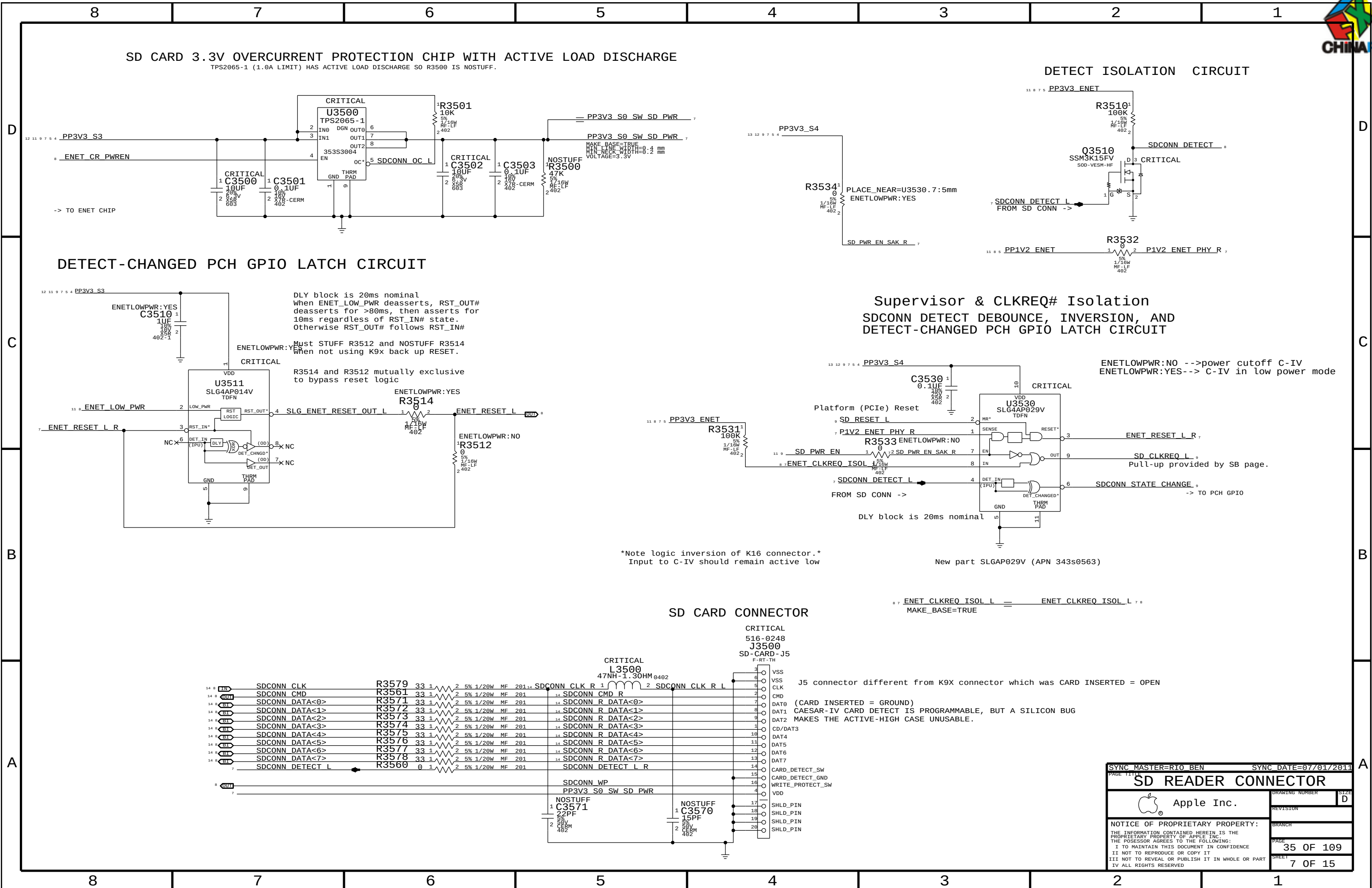


Unused ethernet signals

ENET MDI P<3..0> == NC ENET MDI P<3..0>
ENET MDI N<3..0> == NC ENET MDI N<3..0>
NC ENET WAKE L == NC ENET WAKE L
NC ENET MEDIA SENSE == NC ENET MEDIA SENSE
SYSCLK CLK25M ENET == SYSCLK CLK25M ENET



SYNC MASTER=MASTER		SYNC DATE=MASTER	
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PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
343S0591	1	D2, S7100, SD CNTRL ONLY, B0, QFN 8x8	U3900	CRITICAL	

BCM57765 ENET SR pins are internal 1.2V switching regulator. See note for SR_DISABLE below.
If disabled: Okay to float VDD, VDDP & LX pin. VFB must always connect to =PP1V2_S3_ENET_PHY.
If enabled: VDD/VDDP connect to =PP3V3_S3_ENET_PHY (add bypassing), LX connects to inductor.
Special Star routing needed on these pins. Decoupling on Pg 37.

11 8 7 5 PP3V3 ENET
281mA (1000base-T max power, Caesar IV)

PP1V2 ENET 5 7 8 11
???mA (1000base-T, Caesar V)

From Broadcom:
For SD-Only, can remove C3931, C3930,
C3921, C3920, C3910, C3911, R3910 TBD, R3965 TBD
For SD-Only, RIO removed C3930 as space limited

LR_OUT/GPIO1 is used as a 3.3V/1.8V internal LDO out for
the card reader on-chip I/O.
Connect only to U3900 pin 20.

SDCONN_DETECT use high logic

No MS (Memory Stick) Insert feature needed.
Control signal to light LED or control SD bus power.
SD_DISABLE must be pulled down to use
internal SR. IPD has a race condition.

May not need for SD-Only
Recommended to keep footprint as backup
PHY Non-Volatile Memory
1MBIT ROM contains MAC address, PCIe config
info as well as code for Bonjour proxy.
Required for proper PHY operation.

Caesar IV (ENET) 25MHz Crystal

SYNC MASTER=K91 ERIC		SYNC DATE=10/11/2010	
PAGE TITLE		DRAWING NUMBER	
ETHERNET PHY (CAESAR IV)		SIZE D	
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RIO cable will have pin 1 to pin 1 mapping from rev5!!!

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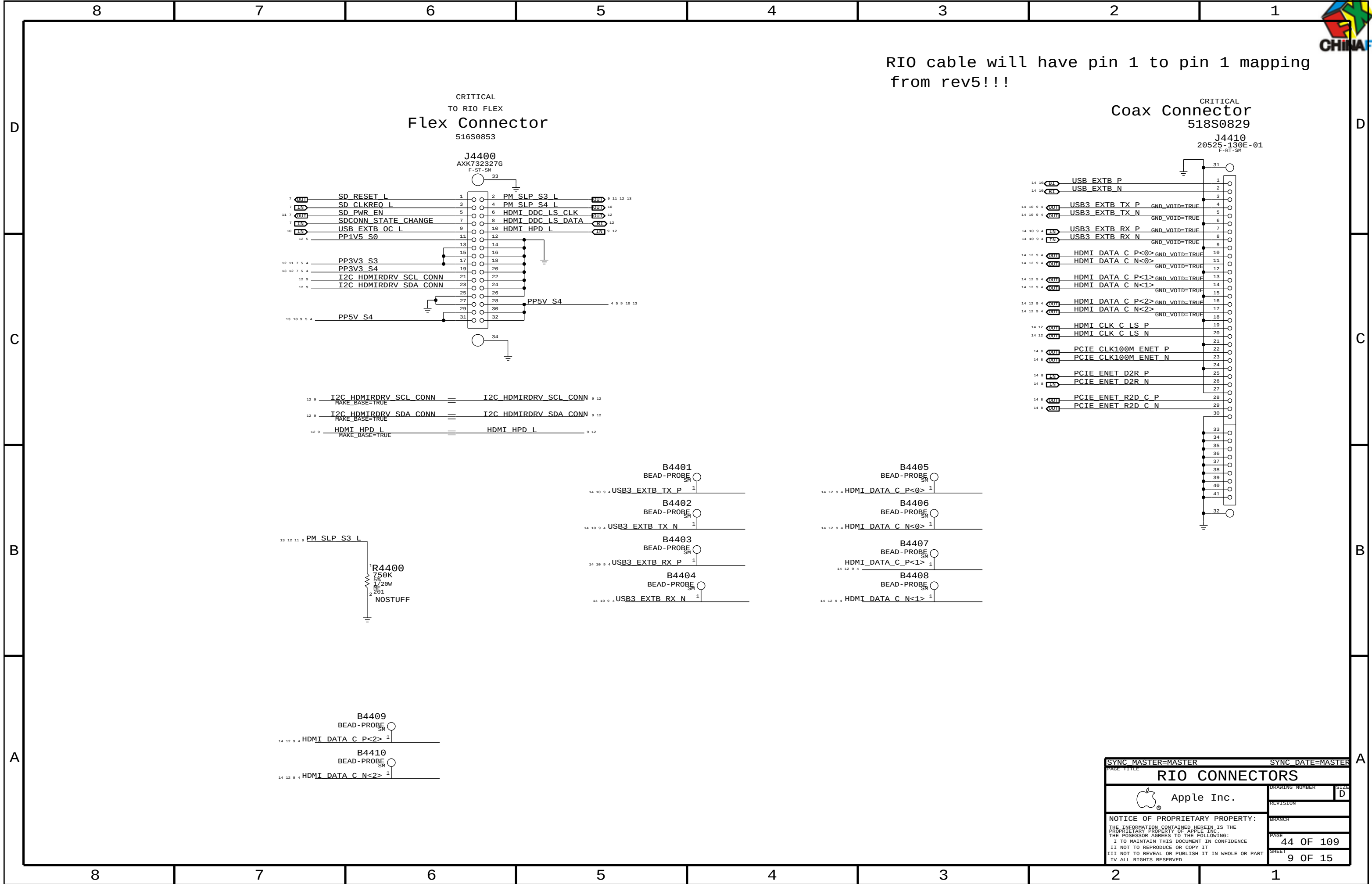
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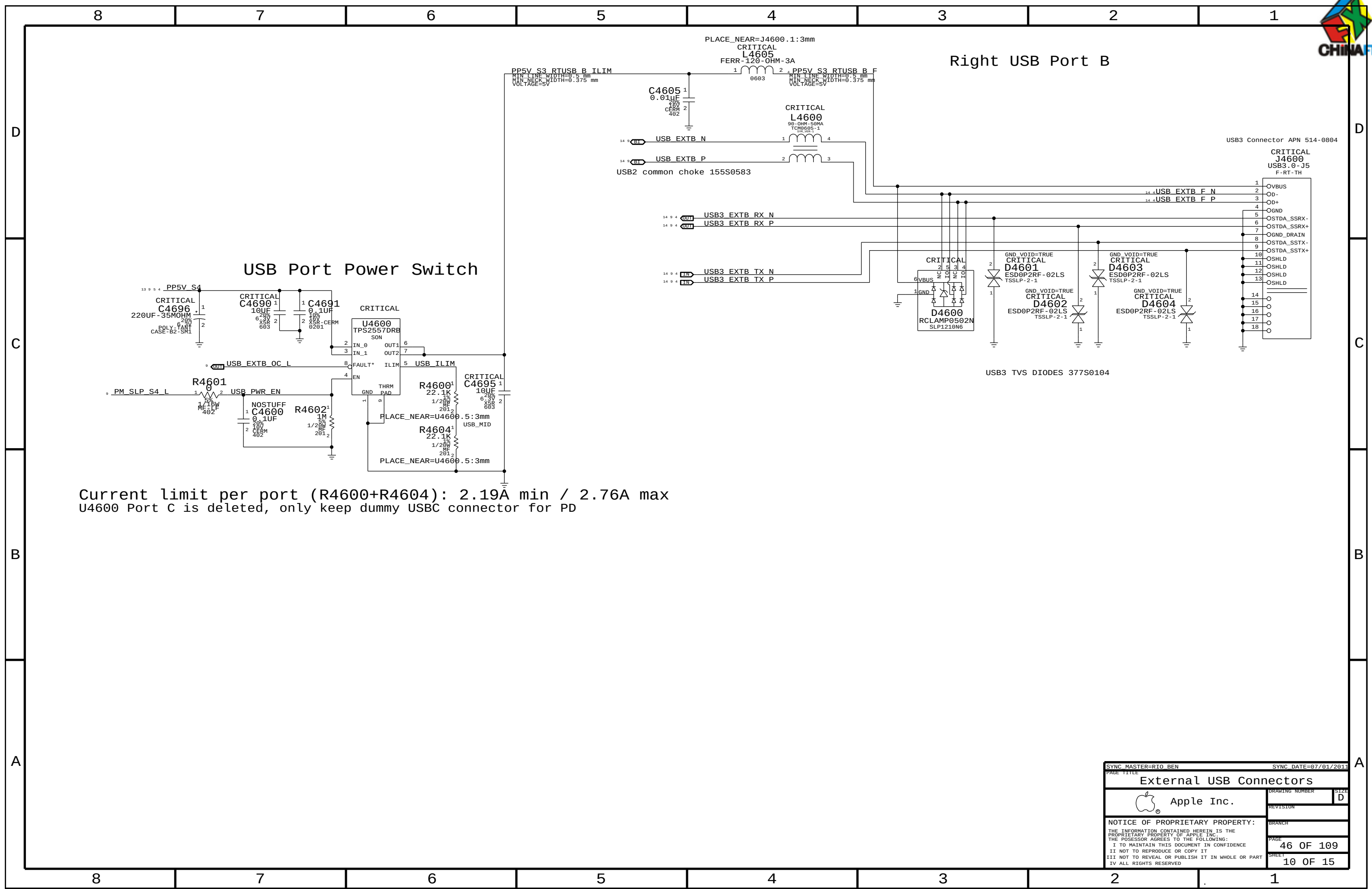
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RIO CONNECTORS		D	
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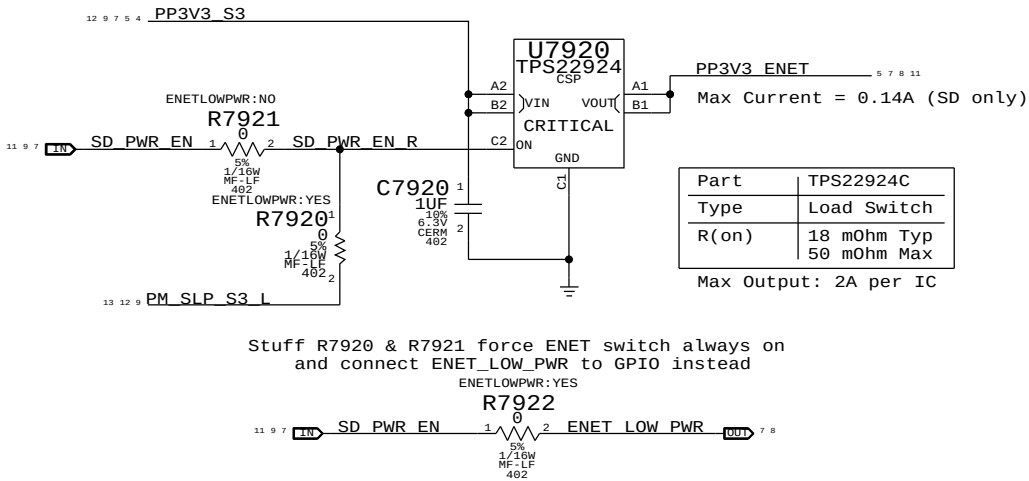
Current limit per port (R4600+R4604): 2.19A min / 2.76A max
U4600 Port C is deleted, only keep dummy USBC connector for PD

SYNC MASTER=RIO BEN		SYNC DATE=07/01/2011	
PAGE TITLE			
External USB Connectors		DRAWING NUMBER	
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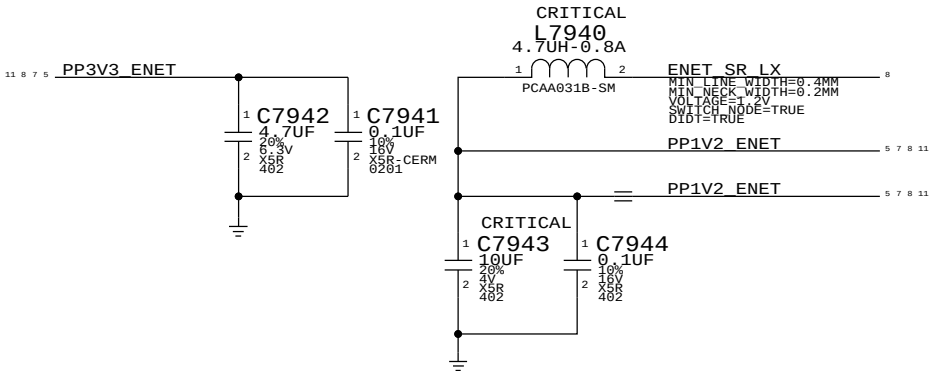


State	SMC_PM_G2_ENABLE	PM_SLP_S5_L	PM_SLP_S4_L	PM_SLP_S3_L
Run (S0)	1	1	1	1
Sleep (S3)	1	1	1	0
Deep Sleep (S4)	1	1	0	0
Deep Sleep (S5)	1	0	0	0
Battery Off (G3Hot)	0	0	0	0

3.3V ENET Switch



CAESAR IV 1.2V INT.VR CMPTS





PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
155S0731	2	FERR BD,60mOHM, 0402	L9700,L9702	CRITICAL	
338S1089	1	HDMI REPEATER - A3	U9700	CRITICAL	

max 1.5V current <400mA
max 3.3V current <30mA
max 1.5V_VDDRX current <180mA
max 1.5V_VDDTX current <200mA
max 1.5V_VDDTA current <20mA

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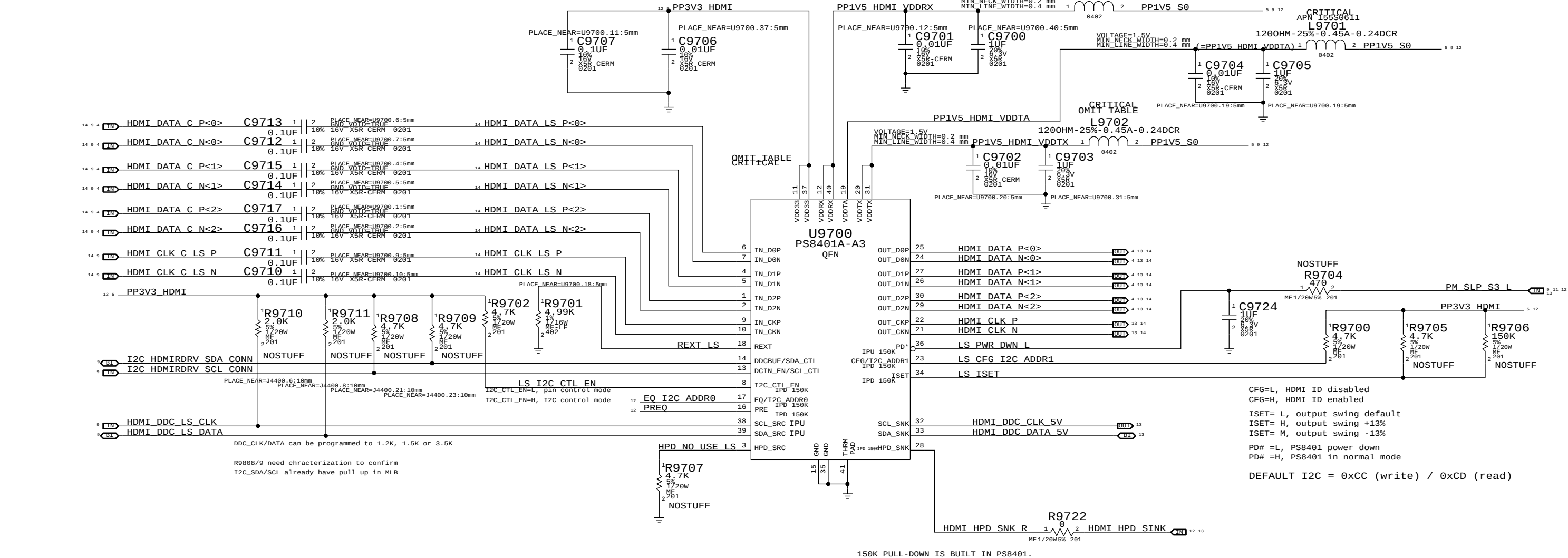
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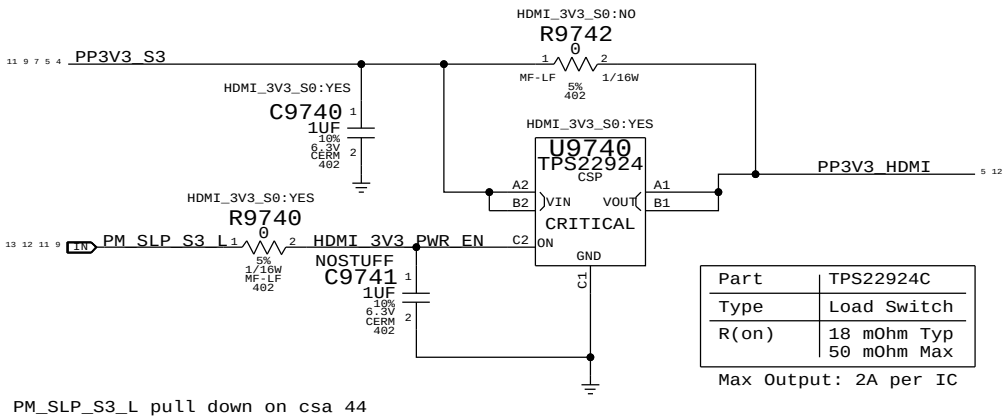
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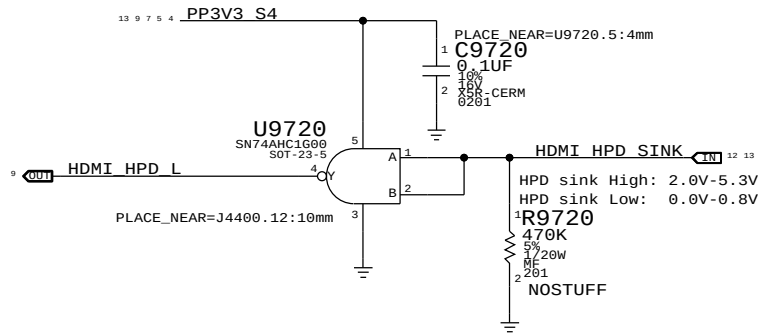
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HDMI 3.3V_S0 Switch



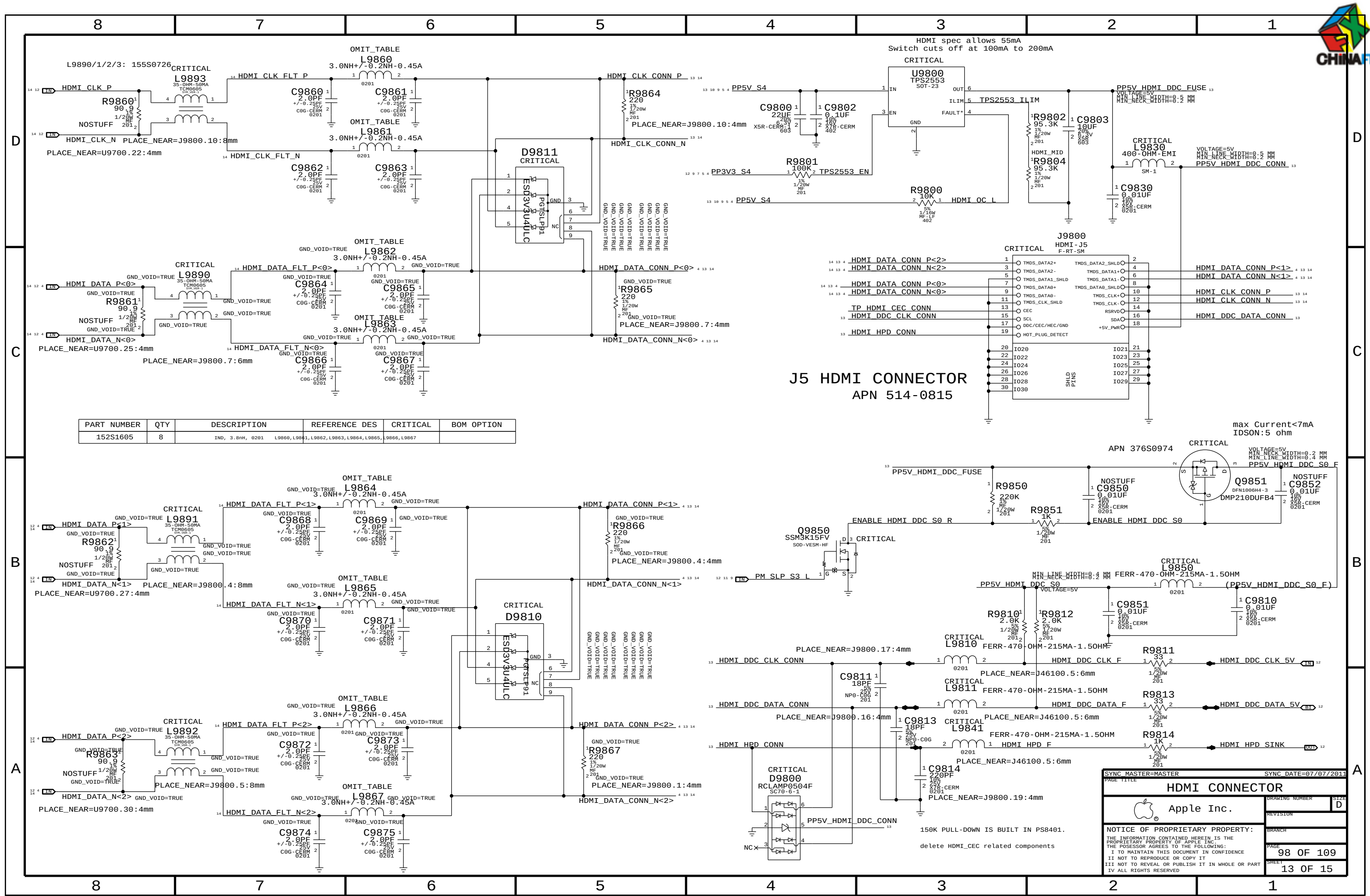
HDMI HPD 5.0V to 3.3V Level Shift



I2C control enabled when R9702 is stuffed

I2C_ADDR1	I2C_ADDR0	I2C ADDRESS(W/R)
L	L	0x4C/4D (default)
L	H	0x5C/5D
H	L	0xCC/CD
H	H	0xEC/ED

SYNC MASTER=MASTER		SYNC DATE=07/07/2011	
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Apple Inc.		DRAWING NUMBER	SIZE D
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PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
152S1605	8	IND, 3.0NH, 0201	L9860, L9861, L9862, L9863, L9864, L9865, L9866, L9867		

SYNC MASTER=MASTER		SYNC DATE=07/07/2011	
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HDMI Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
HDMI_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
HDMI	TOP,BOTTOM	=5X_DIELECTRIC	?
HDMI	ISL3,ISL5,ISL6	=4X_DIELECTRIC	?

SOURCE: AMD/NV CONSTRAINS

PCIE Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
PCIE_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
CLK_PCIE_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
PCIE	TOP,BOTTOM	0.381 MM	?
PCIE	*	0.381 MM	?
CLK_PCIE	*	0.508 MM	?

SOURCE: J5 MLB

USB 2.0 Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
USB_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF
USB_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
USB	TOP,BOTTOM	0.381 MM	?
USB	*	0.381 MM	?

SOURCE: Intel document 467283

USB 3.0 Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
USB3_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF
USB3_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
USB3	TOP,BOTTOM	0.381 MM	?
USB3	ISL3,ISL5,ISL6	=4X_DIELECTRIC	?

SOURCE: Intel document 467283

CAESAR IV (SD) Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
ENET_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
SD_DATA	*	8 MIL	?

SOURCE: K90i

USB Net Properties

NET_TYPE			DIFF PAIR	
ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING		
USB_EXTB	USB_85D	USB	USB_EXTB	USB_EXTB_P 9 10
USB3_EXTB_TX	USB_85D	USB	USB_EXTB	USB_EXTB_N 9 10
USB3_EXTB_TX	USB3_85D	USB3	USB_EXTB_TX	USB3_EXTB_TX_P 4 9 10
USB3_EXTB_TX	USB3_85D	USB3	USB_EXTB_TX	USB3_EXTB_TX_N 4 9 10
USB3_EXTB_RX	USB3_85D	USB3	USB_EXTB_RX	USB3_EXTB_RX_P 4 9 10
USB3_EXTB_RX	USB3_85D	USB3	USB_EXTB_RX	USB3_EXTB_RX_N 4 9 10
USB_EXTB	USB_85D	USB	USB_EXTC	USB_EXTB_F_P 4 10
USB_EXTB	USB_85D	USB	USB_EXTC	USB_EXTB_F_N 4 10
USB3_EXTB_TX	USB3_85D	USB3	USB_EXTC_TX	USB3_EXTB_TX_F_P 4
USB3_EXTB_TX	USB3_85D	USB3	USB_EXTC_TX	USB3_EXTB_TX_F_N 4
USB3_EXTB_RX	USB3_85D	USB3	USB_EXTC_RX	USB3_EXTB_RX_F_P 4
USB3_EXTB_RX	USB3_85D	USB3	USB_EXTC_RX	USB3_EXTB_RX_F_N 4

PCIE Net Properties

NET_TYPE			
ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING	
PCIE_ENET_R2D	PCIE_85D	PCIE	PCIE_ENET_R2D_P 8
PCIE_ENET_R2D	PCIE_85D	PCIE	PCIE_ENET_R2D_N 8
PCIE_ENET_R2D	PCIE_85D	PCIE	PCIE_ENET_R2D_C_P 8 9
PCIE_ENET_R2D	PCIE_85D	PCIE	PCIE_ENET_R2D_C_N 8 9
PCIE_ENET_D2R	PCIE_85D	PCIE	PCIE_ENET_D2R_P 8 9
PCIE_ENET_D2R	PCIE_85D	PCIE	PCIE_ENET_D2R_N 8 9
PCIE_ENET_D2R	PCIE_85D	PCIE	PCIE_ENET_D2R_C_P 8
PCIE_ENET_D2R	PCIE_85D	PCIE	PCIE_ENET_D2R_C_N 8
PCIE_CLK100M	CLK_PCIE_90D	CLK_PCIE	PCIE_CLK100M_ENET_P 8 9
PCIE_CLK100M	CLK_PCIE_90D	CLK_PCIE	PCIE_CLK100M_ENET_N 8 9

HDMI Net Properties

NET_TYPE			
ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING	
HDMT_DATA_LS	HDMT_90D	HDMT	HDMI_DATA_C_P<2..0> 4 9 12
HDMT_DATA_LS	HDMT_90D	HDMT	HDMI_DATA_C_N<2..0> 4 9 12
HDMT_CLK_LS	HDMT_90D	HDMT	HDMI_CLK_C_LS_P 9 12
HDMT_CLK_LS	HDMT_90D	HDMT	HDMI_CLK_C_LS_N 9 12
HDMT_DATA	HDMT_90D	HDMT	HDMI_DATA_P<2..0> 4 12 13
HDMT_DATA	HDMT_90D	HDMT	HDMI_DATA_N<2..0> 4 12 13
HDMT_CLK	HDMT_90D	HDMT	HDMI_CLK_P 12 13
HDMT_CLK	HDMT_90D	HDMT	HDMI_CLK_N 12 13
HDMT_DATA	HDMT_90D	HDMT	HDMI_DATA_CONN_P<2..0> 4 13
HDMT_DATA	HDMT_90D	HDMT	HDMI_DATA_CONN_N<2..0> 4 13
HDMT_CLK	HDMT_90D	HDMT	HDMI_CLK_CONN_P 13
HDMT_CLK	HDMT_90D	HDMT	HDMI_CLK_CONN_N 13
HDMT_DATA	HDMT_90D	HDMT	HDMI_DATA_LS_P<2..0> 12
HDMT_DATA	HDMT_90D	HDMT	HDMI_DATA_LS_N<2..0> 12
HDMT_CLK	HDMT_90D	HDMT	HDMI_CLK_LS_P 12
HDMT_CLK	HDMT_90D	HDMT	HDMI_CLK_LS_N 12
HDMT_DATA	HDMT_90D	HDMT	HDMI_DATA_FLT_P<2..0> 13
HDMT_DATA	HDMT_90D	HDMT	HDMI_DATA_FLT_N<2..0> 13
HDMT_CLK	HDMT_90D	HDMT	HDMI_CLK_FLT_P 13
HDMT_CLK	HDMT_90D	HDMT	HDMI_CLK_FLT_N 13

SD Net Properties

NET_TYPE			
ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING	
CR_DATA	ENET_50S	SD_DATA	SDCONN_DATA<7..0> 7 8
CR_DATA	ENET_50S	SD_DATA	SDCONN_CMD 7 8
CR_CLK	ENET_50S	SD_DATA	SDCONN_CLK 7 8
CR_DATA	ENET_50S	SD_DATA	SDCONN_R_DATA<7..0> 7
CR_DATA	ENET_50S	SD_DATA	SDCONN_CMD_R 7
CR_CLK	ENET_50S	SD_DATA	SDCONN_CLK_R 7
CR_CLK	ENET_50S	SD_DATA	SDCONN_ENET_CLK_R 8

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PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DEFAULT	*	Y	=50_OHM_SE	=50_OHM_SE	10 MM	0 MM	0 MM
STANDARD	*	Y	=DEFAULT	=DEFAULT	10 MM	=DEFAULT	=DEFAULT

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
85_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
85_OHM_DIFF	TOP,BOTTOM	Y	0.112 MM	0.112 MM		0.120 MM	0.120 MM
85_OHM_DIFF	ISL3	Y	0.101 MM	0.101 MM		0.140 MM	0.140 MM
85_OHM_DIFF	ISL5, ISL6	Y	0.118 MM	0.118 MM		0.130 MM	0.130 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
90_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
90_OHM_DIFF	TOP,BOTTOM	Y	0.099 MM	0.099 MM		0.130 MM	0.130 MM
90_OHM_DIFF	ISL3	Y	0.092 MM	0.092 MM		0.160 MM	0.160 MM
90_OHM_DIFF	ISL5, ISL6	Y	0.110 MM	0.110 MM		0.150 MM	0.150 MM

C

B

A

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
1X_DIELECTRIC	TOP, BOTTOM	0.070 MM	?
1X_DIELECTRIC	ISL3	0.120 MM	?
1X_DIELECTRIC	ISL5	0.120 MM	?
1X_DIELECTRIC	ISL6	0.120 MM	?
3X_DIELECTRIC	TOP, BOTTOM	0.210 MM	?
4X_DIELECTRIC	TOP, BOTTOM	0.280 MM	?
5X_DIELECTRIC	TOP, BOTTOM	0.350 MM	?
3X_DIELECTRIC	ISL3, ISL5, ISL6	0.360 MM	?
4X_DIELECTRIC	ISL3, ISL5, ISL6	0.480 MM	?

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